

Problem 18.4

A special perpetuity pays \$200 at the end of each year for the first 10 years and \$100 at the end of each year thereafter. If the effective annual rate of interest is 10%, find the present value of the perpetuity.

Solution

The effective annual interest rate is

$$i = 0.10.$$

The first 10 payments of \$200 form an annuity-immediate, with present value

$$200a_{\overline{10}|}.$$

After year 10, the perpetuity pays \$100 at the end of each year forever. Its value at time 10 is

$$\frac{100}{i}.$$

Discounting this value back to time 0 gives

$$\frac{100}{i}v^{10}.$$

Therefore,

$$PV = 200a_{\overline{10}|} + \frac{100}{i}v^{10}.$$

Using $v = (1.10)^{-1}$,

$$\begin{aligned} PV &= 200 \left(\frac{1 - (1.10)^{-10}}{0.10} \right) + \frac{100}{0.10} (1.10)^{-10} \\ &\approx 1614.4567. \end{aligned}$$

Hence, the present value of the perpetuity is

$$\boxed{\$1,614.46}.$$